Mock Paper 1B — Computer Systems (H446/01)

Time allowed: 2 hours 30 minutes · Total: 140 marks · Answer all questions.

Instructions: Show all working on calculations and logic questions. Mark against the mark scheme at the end.

AO key: **AO1** = knowledge & understanding · **AO2** = application · **AO3** = analysis/evaluation/design.

A running mark tally is shown beside each question. The whole-paper total is verified at the very end.

Questions

Question 1 — Processors, the FDE cycle and performance *(8 marks)*

A games console uses a multi-core processor that supports pipelining.

1(a) During the **decode** and **execute** stages of the fetch–decode–execute cycle, state the role of the **Current Instruction Register (CIR)** and the **Arithmetic Logic Unit (ALU)**. **[2]**

1(b) Explain how **pipelining** can increase the number of instructions completed per second, and state **one** situation in which the pipeline must be flushed. **[3]**

1(c) The processor's clock speed is **3.2 GHz**. State what this value means and explain why simply increasing the clock speed is **not** always the best way to improve performance. **[3]**

(Running total: 8)

Question 2 — Storage *(7 marks)*

2(a) Explain the difference between **primary** and **secondary** storage, giving **one** example of each. **[3]**

2(b) A data centre must archive large volumes of data that are accessed only rarely but must be retained for many years at the lowest possible cost per gigabyte. State the **most appropriate** storage technology and justify your choice. **[2]**

2(c) Explain why **virtual storage** (e.g. cloud storage) is described as a form of secondary storage, and state **one** disadvantage of relying on it. **[2]**

(Running total: 15)

Question 3 — Operating systems, scheduling and interrupts *(9 marks)*

3(a) An operating system uses a **round robin** scheduling algorithm. Explain how round robin scheduling works and state **one** advantage it has over **first come first served (FCFS)**. **[4]**

3(b) Describe how the operating system uses an **interrupt** to handle input arriving from a keyboard while a program is running, referring to the role of the **stack**. **[3]**

3(c) State what is meant by a **device driver** and explain why an operating system needs them. **[2]**

(Running total: 24)

Question 4 — Translators and the stages of compilation *(9 marks)*

4(a) Explain the role of an **assembler** and state why assembly language is described as a **low-level** language. **[3]**

4(b) During compilation, **code generation** and **optimisation** are the final two stages. Describe what happens during each. **[4]**

4(c) A team is developing in a language that is compiled to an intermediate form and run by a virtual machine. Explain **one** advantage and **one** disadvantage of this approach compared with compiling directly to native machine code. **[2]**

(*Running total: 33*)

Question 5 — Software development methodologies *(7 marks)*

5(a) Describe the key stages of the **Waterfall** lifecycle and explain why it is described as a **sequential** model. **[4]**

5(b) A company is developing safety-critical control software for a medical device, where requirements are fixed by regulation and full documentation is required. Justify why a **Waterfall** approach may be more suitable than **Rapid Application Development (RAD)** in this case. **[3]**

(*Running total: 40*)

Question 6 — Object-oriented programming *(9 marks)*

A program models a school's vehicles. A base class `Vehicle` has the private attributes `registration` and `maxSpeed`, and a method `describe()`. A subclass `Minibus` adds the private attribute `seats`.

6(a) Explain what is meant by a **class** and an **object**, using this scenario to illustrate the difference. **[2]**

6(b) Write the class definition for `Vehicle` in pseudocode (or a language of your choice). Include a constructor, private attributes, a public getter for `maxSpeed`, and a `describe()` method that returns a string containing both attributes. **[4]**

6(c) Explain how **inheritance** would be used to define `Minibus`, and state **one** benefit of inheritance to the programmer. **[3]**

(Running total: 49)

Question 7 — Assembly language and addressing modes *(9 marks)*

A processor uses a Little Man Computer (LMC)-style instruction set. Consider this program:

```
        INP
        STA  a
        INP
        STA  b
loop    LDA  b
        BRZ  end
        LDA  total
        ADD  a
        STA  total
        LDA  b
        SUB  one
        STA  b
        BRA  loop
end     LDA  total
        OUT
        HLT
a       DAT
b       DAT
total   DAT  0
one     DAT  1
```

7(a) Trace the program for the inputs **4** then **3** (in that order). State the value output and describe in one sentence the arithmetic operation this program performs. **[5]**

7(b) Explain the difference between **indirect** addressing and **direct** addressing, and state **one** advantage of indirect addressing. **[4]**

(Running total: 58)

Question 8 — Compression, encryption and hashing *(10 marks)*

8(a) Describe how **dictionary-based compression** reduces file size. **[3]**

8(b) Explain what is meant by a **hashing algorithm** and describe how hashing is used to store and check **passwords** securely. **[4]**

8(c) A hash table of size 11 stores keys using the hash function `h(k) = k MOD 11`. The keys **14**, **25** and **36** are inserted in that order. Show the index each key maps to, identify any **collision**, and state one method of resolving it. **[3]**

(Running total: 68)

Question 9 — Databases and SQL *(10 marks)*

A library uses the following tables:

```
Member(MemberID, Surname, JoinDate)
Loan(LoanID, MemberID, ISBN, DueDate, Returned)
```

`Returned` holds the value `'Y'` or `'N'`.

9(a) Explain what is meant by **referential integrity** and describe **one** problem that could occur in the `Loan` table if it were not enforced. **[3]**

9(b) Write an SQL statement to add a new member with `MemberID` **207**, surname **'Okafor'** and `JoinDate` **'2026-06-27'**. **[3]**

9(c) Write an SQL statement that lists the `Surname` of each member together with the `ISBN` and `DueDate` of every loan they have **not yet returned**, sorted by `DueDate` (earliest first). **[4]**

(Running total: 78)

Question 10 — Networks and TCP/IP *(10 marks)*

10(a) Explain the difference between a **MAC address** and an **IP address**, stating at which layer of the TCP/IP stack each is used. **[3]**

10(b) Explain how **protocols** such as **TCP** ensure that data sent across a network arrives **reliably and in the correct order**. **[4]**

10(c) A company hosts a website. Explain the difference between the **client–server** and **peer-to-peer** models, and justify why client–server is appropriate for the company's website. **[3]**

(Running total: 88)

Question 11 — Number representation *(12 marks)*

11(a) Convert the denary number **−53** into **8-bit two's complement**. Show your working and verify your answer. **[3]**

11(b) A floating-point number uses an **8-bit mantissa** and a **4-bit exponent**, both in two's complement, with an assumed binary point after the first mantissa bit. The stored value is:

```
mantissa = 0110 1000   exponent = 0010
```

Calculate the denary value this represents. Show your working, and state whether the number is **normalised**, justifying your answer. **[5]**

11(c) Perform the binary addition `0101 1010 + 0011 0110` in 8 bits. State the 8-bit result and explain whether **overflow** has occurred if the values are interpreted as **signed two's complement**. **[4]**

(Running total: 100)

Question 12 — Data structures *(10 marks)*

12(a) Explain what is meant by a **linked list**, and state **one** advantage and **one** disadvantage of a linked list compared with a static array. **[3]**

12(b) A **binary search tree (BST)** is built by inserting the following keys in order: **50, 30, 70, 20, 40, 60**. Draw the resulting tree and state the order in which the keys are visited by an **in-order traversal**. **[4]**

12(c) State the **Big O** time complexity of: (i) inserting a node at the head of a linked list; (ii) searching a **balanced** binary search tree of *n* nodes; (iii) accessing an element of an array by its index. **[3]**

(Running total: 110)

Question 13 — Boolean algebra and logic circuits *(11 marks)*

13(a) Complete the truth table for the expression `Q = (NOT A) OR (NOT B)` and state the name of the single logic gate to which this expression is equivalent. **[4]**

13(b) Using the laws of Boolean algebra, simplify the expression `Q = A∧(A ∨ B) ∨ A∧C ∨ C` to its simplest form. Show each step and name the law used. **[4]**

13(c) A counter circuit uses a **D-type flip-flop**. State the purpose of a flip-flop in a sequential circuit, and explain why flip-flops are described as the building blocks of computer **memory/registers**. **[3]**

(Running total: 121)

Question 14 — Networking and security *(7 marks)*

14(a) Explain what is meant by **encryption** when transmitting data across a network, and state why **HTTPS** is used in preference to **HTTP** for online banking. **[3]**

14(b) Describe how a **proxy server** can improve both the **performance** and the **security** of a network. **[2]**

14(c) Explain the difference between a **virus** and a **worm**. **[2]**

(Running total: 128)

Question 15 — Extended response: legal, moral and ethical issues *(12 marks)*

A ride-hailing company deploys an **algorithm that automatically decides driver pay and which jobs each driver is offered**. The algorithm uses location data, the driver's acceptance history and customer ratings. Drivers are not told how the algorithm reaches its decisions, and a driver whose rating falls below a threshold is automatically deactivated without human review.

15 Discuss the **legal, ethical and social issues** raised by this automated decision-making system. In your answer you should consider the interests of the **drivers**, the **company** and **society**, and reach a justified conclusion about whether the system should be used in this form. **[12]**

(Running total: 140)

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